Six clues to count

The drops that did not fall.

Name _____ Date _____

All numbers are pretend. Draw, use tens and ones, or show your steps.

1. Little counts 38 dew drops on one leaf and 47 on another. How many drops are there altogether?

38 + 47 = _____

Show your thinking.

2. A spider web has 72 dew beads. After sunrise, 38 evaporate. How many beads remain?

72 − 38 = _____

Show your thinking.

3. A leaf has 28 drops. Then 17 more form, and later 19 evaporate. How many drops remain?

28 + 17 − 19 = _____

Show your thinking.

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4. Little draws 4 rows of dew beads with 6 beads in each row. How many beads does Little draw?

$4 \times 6 = _____$

Show your thinking.

5. Little has drawn 27 drops and wants 60 in the notebook. How many more drops must Little draw?

$27 + ___ = 60$

Show your thinking.

6. Follow Little’s drawing command: add 5 drops each time. The counts are 12, 17, 22. What count comes next?

$12, 17, 22, ___$

Show your thinking.

WONDER DAILY

The drops that did not fall.

A quiet puzzle

Before sunrise, Little Wonderer found tiny drops on a leaf. A spider web beside it wore shining beads. "Did I miss the rain?" Little asked.

The open night

The night had been clear and calm. The leaf gave off heat to the open sky. It cooled and could get colder than the air around it.

Water we cannot see

Air holds water vapor, water in a gas form we cannot see. Air touching the cool leaf cooled too.

Beads on a thread

Tiny drops formed right on the leaf. Water vapor had begun to condense: it changed from a gas into liquid water. That water was dew.

Little's instructions

Little checked the order: cool the leaf, let air touch it, then notice dew. "My commands are like a little program," Little said.

Back into the air

Sunrise warmed the leaf. The beads grew smaller as water began to evaporate: it changed from liquid back into invisible vapor.

Make a little book

1 Fold it into 8 boxes. Open it flat.

2 Fold short end to short end. Cut the thick line. Open it.

3 Fold the long way, pictures out. Push the ends in.

4 Fold into a book. Page 1 goes on the front!

Give the leaf an instruction.

The drops that did not fall. · Curiosity day

Name _____ Date _____

Choose one scene below. Write one command sentence for the leaf's story. You may add a second step.



The open night



Beads on a thread



Back into the air

Draw the leaf before or after your command. Add one label.

leaf · water vapor · condense · dew · evaporate · first · then

First, ... tell the leaf what to do before dew forms

Then, ... tell what to notice on the cool leaf

After sunrise, ... tell what to watch as the leaf warms

Read your sentence. Does it tell what to do, using a command such as Cool, Wait, Look, or Watch?

The drops that did not fall.

01 Can a leaf get wet without rain?



An imagined close look. Drops are enlarged; the picture is not counting data.

Look closer. Look at the drops on the leaf and the nearby spider web.

A little explanation. Yes. Air holds invisible water vapor. Dew forms right on a cool surface when that vapor condenses into liquid drops. It does not fall from the sky like rain.

Try this. Point to where the drops sit. Imagine a drop growing there, instead of falling down.

02 How does a leaf cool during the night?



An imagined close look. Drops are enlarged; the picture is not counting data.

Look closer. The leaf faces a wide, clear sky. The grass is still.

A little explanation. Surfaces give off heat to the sky. On a clear, calm night, a leaf can lose heat and get colder than the air around it. Clouds and wind can slow that cooling.

Try this. Trace the leaf's edge. Say the first command: "Cool the leaf."

03 Where was the water before we saw drops?



An imagined close look. Drops are enlarged; the picture is not counting data.

Look closer. The air above the leaf looks clear. New beads cling to the leaf.

A little explanation. The water was invisible vapor in the air. Air touching the cool leaf cools too. At its dew point, it is cool enough that it cannot keep all its water invisible. Some vapor condenses into liquid dew.

Try this. Imagine clear air touching this leaf. Use "condense" to tell what changed.

04 Where do the drops go after sunrise?



An imagined close look. Drops are enlarged; the picture is not counting data.

Look closer. Sunlight reaches the leaf. Fewer drops remain.

A little explanation. As the sun warms the leaf, liquid water evaporates back into invisible vapor. The water changes form; it does not vanish forever. A cold glass can collect drops on its outside for the same condensation reason: that water comes from the surrounding air.

Try this. Say two commands in order: "Warm the leaf. Watch the drops shrink."